

INVESTMENTS

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CHAPTER 4

Mutual Funds and Other Investment Companies

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Study Guide · Prepared for Personal Use

LO1 · Investment Companies

Most individual investors do not trade securities directly; instead they direct funds to **investment companies** — financial intermediaries that pool investor money and purchase securities on their behalf, then divide claims to those pooled assets among investors proportional to shares held.

Net Asset Value (NAV)

Net Asset Value is the fund's per-share value: the market value of the portfolio's assets, minus liabilities, divided by shares outstanding. NAV is the price at which open-end fund shares are bought and redeemed.

LO2 · Types of Investment Companies

Open-End vs. Closed-End Funds

Open-end funds stand ready to issue or redeem shares at NAV on demand (though a load may apply); the number of shares outstanding changes daily, and the price can never fall below NAV. **Closed-end funds** issue a fixed number of shares that then trade on an exchange like ordinary stock — investors who want out must sell to another investor, not redeem from the fund. As a result, closed-end fund prices routinely diverge from NAV, trading at a premium or discount = $(\text{Price} - \text{NAV})/\text{NAV}$. More funds trade at a discount than a premium, a long-standing puzzle in finance, since liquidating the portfolio would realise full NAV. About \$20 billion was held in TSX-listed closed-end funds as of early 2018.

Other Pooled Investment Vehicles

- **Commingled funds** — partnerships (often run by a bank or insurer) pooling institutional accounts (trusts, pensions) too small individually to manage separately; bought and sold at NAV like open-end funds, but via units rather than shares.
- **REITs** — real estate investment trusts, similar in structure to closed-end funds. Equity trusts (the majority of Canadian REITs) own property directly and distribute rental/leasing income; mortgage trusts invest in mortgage and construction loans instead.
- **Hedge funds** — privately structured pools, open only to wealthy/institutional investors, subject to minimal regulation. Often impose multi-year lock-ups so managers can hold illiquid assets without redemption pressure. Can use leverage, derivatives, and short sales freely. Grew from about US\$50 billion (1990) to roughly US\$3 trillion (2016) in assets under management.

LO3 · Mutual Funds

Mutual funds are the dominant vehicle for individual investors. Over 4.9 million Canadian households owned mutual funds in 2015; industry assets under management reached almost **\$1.5 trillion** by end-2017 (up from about \$460 million in 2000), dipping only in 2008 due to the financial crisis. **Management companies** (e.g., CI Financial, Invesco, Fidelity, and the major banks' asset-management arms) run entire families of funds under one umbrella, simplifying record-keeping and letting investors switch between funds easily.

Classification by Investment Policy

Fund Type	Key Features
Money Market Funds	Hold short-term instruments (commercial paper, repos, CDs); average maturity just over 1 month. NAV is fixed at \$10/share in Canada (\$1 in the U.S.) to avoid capital gain/loss tax complications on redemption. In September 2008, the U.S. Reserve Primary Fund famously "broke the buck," with NAV falling to \$0.97 after Lehman commercial paper losses.
Equity Funds	Invest primarily in stock, typically holding 4–5% in money market instruments for redemption liquidity. Income funds favour high dividend-yield stocks; growth funds forgo current income for capital-gain potential and carry more risk.
Sector Funds	Concentrate on one industry (energy, financials, technology) or one country/region.
Bond Funds	Specialise in fixed income — further divided by issuer (government/corporate), maturity (short/intermediate/long), or credit quality (investment-grade to high-yield "junk").
International Funds	Global funds invest worldwide including Canada/U.S.; international funds exclude Canada/U.S.; regional funds target one part of the world; emerging-market funds target developing economies.
Balanced / Life-Cycle Funds	Hold both equities and fixed income in relatively stable proportions. Life-cycle funds range from aggressive (younger investors) to conservative (older investors); target-maturity variants automatically grow more conservative as a target date approaches.

LO4 · Costs of Investing in Mutual Funds

Choosing a fund requires looking beyond stated policy and past performance to its fee structure. Four general categories of fees apply:

Fee Type	How It Works
Management Expense Ratio (MER)	Covers administrative expenses, marketing, taxes, and advisory fees, expressed as a % of assets under management. Deducted continuously from fund assets (never billed explicitly). 2015 Canadian averages: 2.35% for equity funds, 1.51% for fixed-income funds, 2.16% for asset-allocation funds, 0.60% for money market funds.
Front-End Load (ISC)	A sales commission paid when shares are purchased, capped at 5% (low-load funds: up to 3%; no-load funds: 0%). Reduces the amount actually invested — e.g., a 6% load on \$1,000 leaves only \$940 invested, requiring a 6.4% return on that net amount just to break even.
Back-End Load (DSC)	A deferred/contingent redemption fee paid on exit, typically starting at 5–6% and falling by 1 percentage point per year held (often to zero by year 5–6).
Trailing Commissions	Ongoing "trailer fees" paid to the selling broker, deducted from fund assets and folded into the MER — never billed explicitly to the investor.

Share Series

Many funds offer multiple "series" of the same underlying portfolio with different fee structures: Series A (various load options), Series D (lower trailers, sold via discount brokers), Series F (fee-based accounts, no trailer since the advisor is paid directly), and institutional series (lower MER, higher minimum investment).

Soft Dollars and the Trading Expense Ratio (TER)

Managers can earn **soft-dollar credits** by directing trades to a particular broker, who then covers some fund expenses (databases, research) in exchange. This can make a fund's reported MER look artificially low, while the true cost surfaces as higher trading commissions embedded in returns rather than in fees. Canadian funds are required to separately report a **Trading Expense Ratio (TER)** — trading commissions as a % of assets — for transparency.

LO5 · Taxation of Mutual Fund Income

Under Canadian tax law, a mutual fund itself generally pays no tax; investment income (interest, dividends, capital gains) flows through and is taxed **in the hands of the investor**, provided the fund distributes substantially all of its income annually. A key trap: because capital gain/dividend distributions are typically paid out once or twice a year to *all* current unitholders, an investor who buys shares **shortly before** a distribution can be taxed on gains realised by the fund *before* that investor even owned the shares — a particular concern late in the calendar year when such distributions are common.

LO6 · Exchange-Traded Funds (ETFs)

ETFs let investors trade index portfolios exactly like shares of stock, unlike mutual funds, which transact only once daily at NAV. Canada launched the **first ETF in the world in 1990**: the Toronto Index Participation Fund (TIPS), tracking the top 35 TSX stocks (later merged into today's iShares S&P/TSX 60 Index ETF). The first U.S. ETF followed in 1993: the SPDR ("spider"), tracking the S&P 500.

ETF Name	Ticker	Index Tracked
iShares S&P/TSX 60 Index	XIU	S&P/TSX 60 Index
iShares Core S&P/TSX Capped Composite	XIC	S&P/TSX Capped Composite Index
SPDR ("Spiders")	SPY	S&P 500
Diamonds	DIA	Dow Jones Industrials
Cubes	QQQ	NASDAQ 100
Vanguard Total Stock Market	VTI	Wilshire 5000

Key insight: ETFs combine index-fund-like low costs with stock-like intraday tradability, and generally offer better tax efficiency than mutual funds because in-kind share creation/redemption limits triggering embedded capital gains for existing holders.

LO7 · Mutual Fund Investment Performance: A First Look

A natural benchmark question: how does actively managed equity fund performance compare with simply holding a passive index? Using the Wilshire 5000 as a benchmark, the average diversified equity fund **underperformed** the index in 27 of 45 years from 1971 to 2015. The index averaged 12.1% annually, about **1 percentage point higher** than the average mutual fund — and this gap persists even after allowing for the small costs of running an index fund (~0.10–0.15% expense ratio).

This finding motivates a deeper question, explored fully in Chapter 11 on the efficient market hypothesis: is underperformance evidence that active management adds no value on average, or could skilled managers still be identified in advance? Researchers test this by examining **performance persistence** — do funds that rank in the top half of returns in one period tend to rank in the top half again the next period? The evidence is mixed: some persistence appears in short-term studies, but it is inconsistent and can partly reflect differences in cost structure or fund style rather than genuine manager skill.

Why this matters for LO4: expenses are one of the few things about a fund an investor can predict with confidence in advance. Since average returns are frequently roughly in line with the market (or slightly below), the surest way to add value is often to minimise costs — a central argument for indexing.

LO8 · Information on Mutual Funds

Investors evaluating funds rely on standardised third-party data providers rather than raw fund marketing material. **Morningstar** (morningstar.ca) is the primary Canadian source, reporting NAV, day change, trailing-twelve-month yield, total assets, expense ratios, historical performance, and Morningstar's proprietary star ratings for risk-adjusted performance versus peers.

Every fund's **prospectus** legally discloses its investment policy, fee structure, and risks — the essential starting point before comparing a fund's marketing claims to independent data. Investors should always cross-reference a fund's stated MER, load structure, and TER (Trading Expense Ratio) against Morningstar or similar sources rather than relying solely on the fund company's own materials.

Chapter 4 — Big-Picture Takeaways

- Investment companies pool investor funds and divide ownership via NAV; open-end funds transact at NAV on demand, while closed-end funds trade on exchanges and often diverge from NAV.
- Fees compound powerfully over time — small differences in MER produce large differences in ending wealth over long horizons (Table 4.1).
- Canadian mutual fund income flows through to investors for tax purposes; timing of purchases around distribution dates matters.
- ETFs (a Canadian invention, 1990) combine index-fund cost efficiency with stock-like intraday tradability and generally better tax efficiency than mutual funds.
- On average, actively managed equity funds have historically underperformed passive benchmarks by about 1% annually, and outperformance shows little consistent persistence from year to year.
- Third-party data (Morningstar) and the fund prospectus are the essential tools for evaluating any specific fund before investing.

NAV and the impact of fees on net returns are the core quantitative skills tested from this chapter.

F1 | Net Asset Value (NAV)

$$\text{NAV} = (\text{Market Value of Assets} - \text{Liabilities}) / \text{Shares Outstanding}$$

The fund's per-share value. Open-end funds transact at this price (plus any load); closed-end fund market prices can differ from NAV.

F2 | Closed-End Fund Premium/Discount

$$\text{Premium (Discount)} = (\text{Price} - \text{NAV}) / \text{NAV}$$

A negative result is a discount (price below NAV); positive is a premium. Most closed-end funds trade at a discount — a long-standing empirical puzzle.

F3 | Rate of Return Net of Expenses

$$\text{Net Return} = \text{Gross Portfolio Return} - \text{Expense Ratio (MER)}$$

Approximation that holds well when the expense ratio is deducted from asset value over the period. E.g., a 10% gross return with a 1% MER yields approximately a 9% net return to the investor.

F4 | Front-End Load Break-Even Return

$$\text{Net Investment} = \text{Amount Paid} \times (1 - \text{Load \%})$$

$$\text{Break-even Return} = \text{Load \%} / (1 - \text{Load \%})$$

A 6% load on \$1,000 leaves \$940 invested. Break-even = $60/940 = 6.4\%$ cumulative return on the net investment just to recover the load's cost.

F5 | Back-End (Deferred) Load Schedule

$$\text{DSC}(\text{year } n) = \text{Initial DSC \%} - (n - 1) \times 1 \text{ percentage point, floor at } 0\%$$

Typical Canadian pattern: starts at 5–6% in year 1, falls by 1 point per full year held, reaching 0% around year 5–6.

EXAMPLE 4.1 | Net Asset Value

Given: Portfolio worth \$120 million; fund owes \$4 million to investment advisers and \$1 million in rent/wages/misc.; 5 million shares outstanding.

$$\text{NAV} = (\$120\text{M} - \$4\text{M} - \$1\text{M}) / 5\text{M shares} = \$115\text{M} / 5\text{M} = \$23.00 \text{ per share}$$

EXAMPLE 4.3 | Fees and Net Returns

Given: Fund starts the year with \$100 million in assets, 10 million shares outstanding. Underlying portfolio grows 10% (no income). MER = 1%.

$$\text{Initial NAV} = \$100\text{M} / 10\text{M} = \$10.00/\text{share}$$

$$\text{Gross growth: } \$100\text{M} \times 1.10 = \$110\text{M} \rightarrow \text{Gross NAV} = \$11.00/\text{share} \text{ (10\% gross return)}$$

$$\text{MER deduction: } 1\% \times \$110\text{M} \approx \$1\text{M} \rightarrow \text{Net assets} = \$109\text{M} \rightarrow \text{Net NAV} = \$10.90/\text{share}$$

$$\text{Net return} = (\$10.90 - \$10.00)/\$10.00 = 9.0\% = \text{Gross return (10\%)} - \text{MER (1\%)}$$

Table 4.1 — Compounding the Cost of Fees: \$10,000 invested at 12% gross return annually, across three fee structures.

Horizon	Fund A (0.5% MER, no load)	Fund B (1.5% MER, no load)	Fund C (1% MER, 8% load)
Initial	\$10,000	\$10,000	\$9,200
5 years	\$17,234	\$16,474	\$15,502
10 years	\$29,699	\$27,141	\$26,123
15 years	\$51,183	\$44,713	\$44,018
20 years	\$88,206	\$73,662	\$74,173

Notice: even a 1-percentage-point difference in annual MER (Fund A vs. B) compounds to a ~20% difference in ending wealth over 20 years. Fund C's upfront load costs less over very long horizons than Fund B's higher ongoing MER — the drag from an annual fee compounds, while a load is a one-time hit.

PRACTICE PROBLEM | Chinook Asset Management**Scenario**

Chinook Asset Management, based in Calgary, runs an equity mutual fund. At the start of the year, the fund holds a portfolio worth \$250 million, owes \$8 million to its investment advisers and \$2 million in operating expenses, and has 12 million shares outstanding.

Part A — Net Asset Value

- (i) Calculate the fund's NAV per share at the start of the year.
- (ii) The fund also runs a closed-end sibling fund with the same NAV, currently trading at \$18.50 on the TSX. Calculate its premium or discount to NAV.
- (iii) Explain in one or two sentences why closed-end fund prices can diverge from NAV even though open-end fund prices cannot.

Part B — Fees and Net Return

Over the year, Chinook's underlying portfolio (before fees) grows by 14%. The fund's MER is 2.1%.

- (i) Calculate the fund's approximate net return to investors.
- (ii) A client invests \$10,000 in a Series A unit with a 5% front-end load. How much is actually invested after the load?
- (iii) What cumulative return on the net investment is required just to break even on the 5% load?

Part C — Comparing Fee Structures

A second client is deciding between two share classes of the same Chinook fund, both expected to earn 11% gross annually for 15 years on a \$20,000 investment:

Series A: 3% front-end load, 1.8% MER. Series D: no load, 1.0% MER (discount-broker sold).

- (i) Estimate the approximate ending value for each series after 15 years (you may assume the MER reduces the annual return each year, compounded).
- (ii) Which series leaves the client better off, and by roughly how much?
- (iii) Under what circumstance (very short holding period) might the higher-load, lower-MER series ever be preferable? Explain the trade-off conceptually.

Part D — Fund Type & Structure

Chinook is considering launching three new products: (1) a fund tracking the S&P/TSX 60 that trades intraday on the TSX; (2) a fund investing exclusively in Alberta energy-sector equities; (3) a private, lightly-regulated pool open only to institutional investors using leverage and derivatives.

- (i) Classify each of the three products using the terminology from this chapter (e.g., ETF, sector fund, hedge fund).

- (ii) For product (1), briefly explain why Canada has a particular claim to have pioneered this fund structure.
- (iii) For product (3), explain one regulatory reason its structure allows strategies not available to a standard mutual fund.

Hints & Guidance

Part A tests LO1–LO2 and Formulas F1–F2. Part B tests LO4–LO5 and Formulas F3–F4 (mirror Example 4.3 exactly). Part C tests LO4, modelling Table 4.1's logic with new numbers — compound the net-of-MER return annually, then subtract the load once at the start. Part D synthesises LO2 and LO6.